Ch 5.2: The Bootstrap

Lecture 16 - CMSE 381

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Announcements

Last time:

- k-fold CV for Classification

Announcements:

- Fifth homework posted

Due Weds
10/18

Covered in this lecture

- Bootstrap

Section 1

The Bootstrap

The Idea

The goal: quantify the uncertainty associated with a given estimator or statistical learning method.

↳ The Bootstrap

Can be used in many stat settings

Today: Bootstrap by example

HW: Bootstrap to analyze linear regression

The boostrap idiom

"Pull yourself up by your bootstraps."

- Originally used as a saying meaning an impossible task, used sarcastically
- Now often used to imply that socioeconomic advancement is something everyone should be able to do

· Also the origin of "the term booting a computer"

Today's class: Bootstrap on a simple modeling problem

α

- We wish to invest a fixed sum of money in two financial assets that yield returns of X and Y, respectively, where X and Y are random quantities.

Not input variables or output variables just happens to be X, Y

- We will invest a fraction α of our money in X, and will invest the remaining $1 - \alpha$ in Y.

% of my money

$\alpha \in [0,1]$

- Since there is variability associated with the returns on these two assets, we wish to choose α to minimize the total risk, or variance, of our investment.

Return: $\alpha X + (1-\alpha) Y$

Min: $Var(\alpha X + (1-\alpha) Y)$

One can show.....

...that $\mathrm{Var}(\alpha X + (1-\alpha)Y)$
is minimzed by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

where

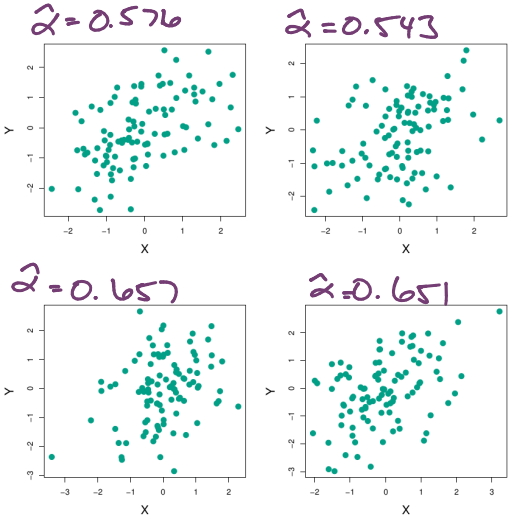
- $\sigma_X^2 = \mathrm{Var}(X)$
- $\sigma_Y^2 = \mathrm{Var}(Y)$
- $\sigma_{XY} = \mathrm{Cov}(X, Y)$

Get an estimate:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}$$

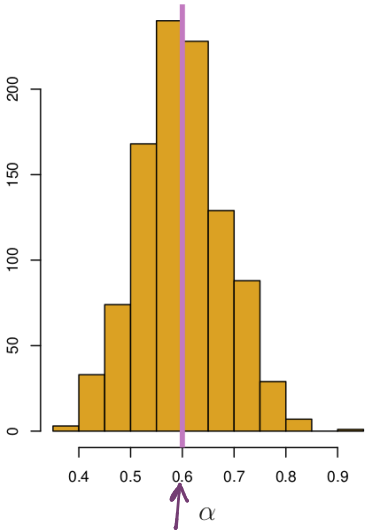
Calculate from data

Simulated data



- Simulate data using
  - $\sigma_X^2 = 1$
  - $\sigma_Y^2 = 1.25$
  - $\sigma_{XY} = 0.5$
  - $\boxed{\text{Implies } \alpha = 0.6}$
- In each panel: Simulate 100 pairs of returns for investments
- Predict $\hat{\sigma}_X^2$, $\hat{\sigma}_Y^2$, and $\hat{\sigma}_{XY}$
- $\hat{\alpha}$ prediction by panel:
  0.576 0.543
  0.657 0.651

Resimulate: Rinse and repeat



- Simulate 100 data points 1,000 times
- Left: Histogram of ~~predictions for~~ $\hat{\alpha}$
- Pink line: True value for α
- Mean over simulated values: $0.5996 = \bar{\alpha} = \frac{1}{1000}\sum \hat{\alpha}_r$
- St ~~dev~~ Error: $0.083 = \sqrt{\frac{1}{1000-1}\sum(\hat{\alpha}_r - \bar{\alpha})}$

Coding part 1: Apprximate $\hat{\alpha}$ with simulated data

So what's the problem?

I can't simulate my data.

Bootstrap:

Create simulations from my given data set

The solution: Sample the data with replacement

↳ can end up with the same data point multiple times

n=3

" 1st Bootstrap sample

Obs	X	Y
1	4.3	2.4
2	2.1	1.1
3	5.3	2.8

Original Data (Z)

Z^{*1}

Obs	X	Y
3	5.3	2.8
1	4.3	2.4
3	5.3	2.8

$\hat{\alpha}^{*1}$

2nd

Z^{*2}

Obs	X	Y
2	2.1	1.1
3	5.3	2.8
1	4.3	2.4

$\hat{\alpha}^{*2}$

Z^{*B}

Bth Bootstrap sample

Obs	X	Y
2	2.1	1.1
2	2.1	1.1
1	4.3	2.4

$\hat{\alpha}^{*B}$

Sample data set has n points (same as my original)

Computation of error

- Repeate procedure B times:
- Get B bootstrap data sets, $Z^{*1}, Z^{*2}, \ldots, Z^{*B}$
- Get B bootstrap estimates $\hat{\alpha}^{*1}, \hat{\alpha}^{*2}, \ldots, \hat{\alpha}^{*B}$

Get standard error estimate:

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \underbrace{\frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}}_{\overline{\hat{\alpha}^{*}}}\right)^2}$$

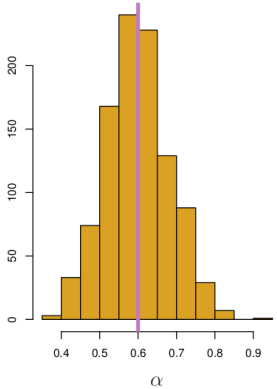
Standard deviation of [$\hat{\alpha}$] list

each Bootstrap estimate

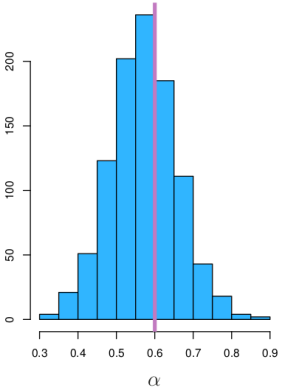
But denominator is B−1

Coding part 2: Resampling data

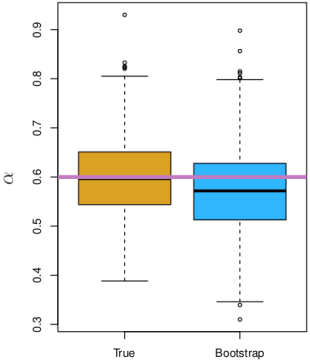
Back to the example



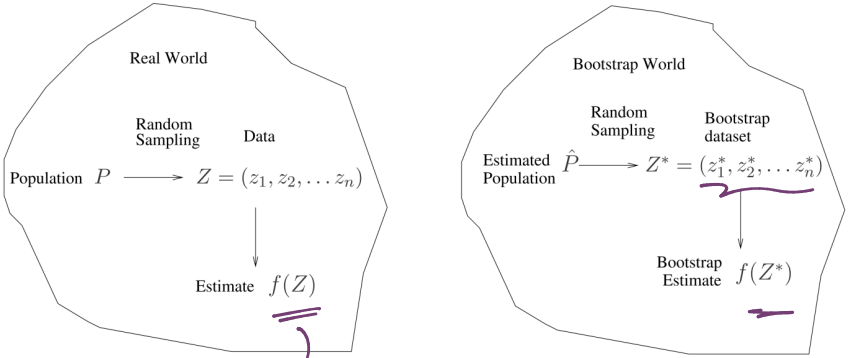
Resample version
Predicted $SE(\hat{\alpha}) = 0.083$



Bootstrap version
Predicted $SE(\hat{\alpha}) = 0.087$



A general picture for the bootstrap



my 2 in this example

TL;DR

- Start with data set of n points
- Sample n points with replacement to get data set Z^{*1}
- Use this to estimate whatever parameter we want $\hat{T}^{*1}$

- Repeat B times to get estimates $\hat{T}^{*1}, \ldots, \hat{T}^{*B}$
- Estimate standard error of our T estimate by

$$SE_B(\hat{T}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{T}^{*r} - \frac{1}{B}\sum_{r'=1}^{B}\hat{T}^{*r'}\right)^2}$$

2 standard deviation

Boostrap vs Cross-Validation

Bootstrap:

Taking sample with replacement

Size same as n

Randomness from doing this B times

(error)

Goal: establish emperical dist. function for some statistic

CV:

Taking subset without replacement!

Subset < n

k-fold CV randomness comes from shuffling but then subsetting parts

Goal: measuring performance of a model

Next time

12	Mon	Oct 2	Leave one out CV	5.1.1, 5.1.2	
13	Wed	Oct 4	k-fold CV	5.1.3	
14	Fri	Oct 6	More k-fold CV,	5.1.4-5	
15	Mon	Oct 9	k-fold CV for classification	5.1.5	HW #4 Due
16	Wed	Oct 11	Resampling methods: Bootstrap	5.2	
17	Fri	Oct 13	Subset selection	6.1	
18	Mon	Oct 16	Shrinkage: Ridge	6.2.1	
19	Wed	Oct 18	Shrinkage: Lasso	6.2.2	HW #5
	Fri	Oct 20	**Review**		
	Mon	Oct 23	No class - Fall break		
	Wed	Oct 25	**Midterm #2**		
20	Fri	Oct 27	Dimension Reduction	6.3	